

HDOC Day_23 Programming In C

Problem

Write a C program that displays the following menu and uses a switch statement to print the basic

square/rectangular pattern selected by the user. Use nested loops without arrays. Each pattern may be implemented

in a separate parameterless function with no return value.

Note: For each activity, prepare the algorithm and test-case table. Then write, compile and execute the C program,

and test it against the prepared test cases.

1. Solid square pattern

Input the number of rows, n, and print a solid square pattern containing n rows and n columns.

For n = 4:

```
* * * *
```

```
* * * *
```

```
* * * *
```

```
* * * *
```

2. Solid rectangular pattern

Input the number of rows, m, and columns, n, and print a solid rectangular pattern.

For m = 3 and n = 5:

```
* * * * *
```

```
* * * * *
```

```
* * * * *
```

3. Row-number pattern

Input the number of rows, m, and columns, n. Print a rectangular number pattern in which every element in a

row is equal to its row number.

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For m = 4 and n = 4:

1 1 1 1

2 2 2 2

3 3 3 3

4 4 4 4

4. Column-number pattern

Input the number of rows, m, and columns, n. Print a rectangular number pattern in which every element in a column is equal to its column number.

For m = 4 and n = 4:

1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

ALGORITHM

Main Algorithm

1. **Start**
2. **Display Menu:**
 - Print option 1: Solid Square Pattern
 - Print option 2: Solid Rectangular Pattern
 - Print option 3: Row Number Pattern
 - Print option 4: Column Number Pattern
3. **Read** user choice into ch.
4. **Evaluate ch using Switch Case:**
 - **If ch == 1:** Call Function solid_square().
 - **If ch == 2:** Call Function solid_rectangular().
 - **If ch == 3:** Call Function row_number().
 - **If ch == 4:** Call Function column_number().

- **Default:** Print "invalid input".

5. Stop

Function Algorithms

- **Algorithm for solid_square()**
 1. Prompt user and **Read** number of sides into n.
 2. For i from 1 to n (Outer loop for rows):
 - For j from 1 to n (Inner loop for columns):
 - Print "*" "
 - Print a newline (\n).
 3. Return to caller.
- **Algorithm for solid_rectangular()**
 1. Prompt user and **Read** rows into n and columns into m.
 2. For i from 1 to n:
 - For j from 1 to m:
 - Print "*" "
 - Print a newline (\n).
 3. Return to caller.
- **Algorithm for row_number()**
 1. Prompt user and **Read** rows into n and columns into m.
 2. For i from 1 to n:
 - For j from 1 to m:
 - Print value of i followed by space.
 - Print a newline (\n).
 3. Return to caller.
- **Algorithm for column_number()**
 1. Prompt user and **Read** rows into n and columns into m.
 2. For i from 1 to n:
 - For j from 1 to m:

- Print value of j followed by space.
 - Print a newline (\n).
3. Return to caller.

C PROGRAM CODE

```
#include <stdio.h>

void solid_square()
{
    int n;

    printf("enter no of rows and columns(both are equal)\n");
    scanf("%d",&n);
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=n;j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}

void solid_rectangular()
{
    int n,m;

    printf("enter no of rows and coloumns\n");
    scanf("%d%d",&n,&m);
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
```

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```
{
    printf("* ");
}
printf("\n");
}
}
void row_number()
{
    int n,m;
    printf("enter no of rows and coloumn\n");
    scanf("%d%d",&n,&m);
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
        {
            printf("%d ",i);
        }
        printf("\n");
    }
}
void column_number()
{
    int n,m;
    printf("enter no of rows and coloumn\n");
    scanf("%d%d",&n,&m);
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=m;j++)
```

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```
{
    printf("%d ",j);
}
printf("\n");
}
}
int main()
{
    int ch;
    printf("choice 1 is to fill a solid square pattern\n");
    printf("choice 2 is to print a solid rectangular pattern\n");
    printf("choice 3 is to print a row number pattern\n");
    printf("choice 4 is to print a column number pattern\n");
    printf("enter your choice\n");
    scanf("%d",&ch);
    switch (ch)
    {
        case 1:
        {
            solid_square();
            break;
        }
        case 2:
        {
            solid_rectangular();
            break;
        }
        case 3:
```

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```
{
    row_number();
    break;
}
case 4:
{
    column_number();
    break;
}
default:
{
    printf("invalid input\n");
    break;
}
}
return 0;
}
```